

Lecture 3

Describing and Summarizing Data

Part 1.

INTRODUCTION TO DESCRIPTIVE STATISTICS

| Making Sense of Data

Data by themselves rarely tell a story.

Descriptive statistics provides the tools to organize, summarize, visualize, and communicate information contained in data.

It helps us transform raw observations into clear, meaningful information through:

- Summarization
- Visualization
- Interpretation

To do this, we first need to understand the fundamental building block of descriptive statistics: **the statistic**.

| What is a Statistic?

A **statistic** is a numerical summary computed from a sample.

Examples include the sample mean, median, standard deviation, correlation, minimum, etc.

A statistic summarizes an important aspect of the data and helps answer a specific question.

Example

"The standard deviation of daily returns over the last three years is 1.2% per day."

A statistic is meaningful only when it is **interpreted in context**. Whenever you encounter a reported statistic, ask:

- **Which variable** does it describe?
- **Which sample** was used?
- **Over what period?**
- **What question** does it answer?

Fundamental Concepts

Before computing statistics, we need to understand the structure of a dataset.

A dataset is built from four fundamental concepts:

- ❑ **Population:** the complete group or process we want to study.
- ❑ **Sample:** the subset of observations we actually collect.
- ❑ **Observation:** one individual unit in the dataset.
- ❑ **Variable:** one characteristic measured for each observation.

In a standard data table, **rows represent observations** and **columns represent variables**.

Statistic, Parameter, Estimator, Estimate

These four terms are closely related but describe different ideas.

- ❑ A **parameter** is the (usually unknown) numerical characteristic of the entire population.
- ❑ A **statistic** is any number computed from a sample.
- ❑ An **estimator** is the rule or formula used to estimate a parameter.
- ❑ An **estimate** is the numerical value obtained when applying that estimator to one particular sample.

For example, the **sample mean** is an estimator of the **population mean**. The value computed from one sample, such as **0.7% per month**, is the estimate.

A different sample would generally produce a different estimate, even when using the same estimator.

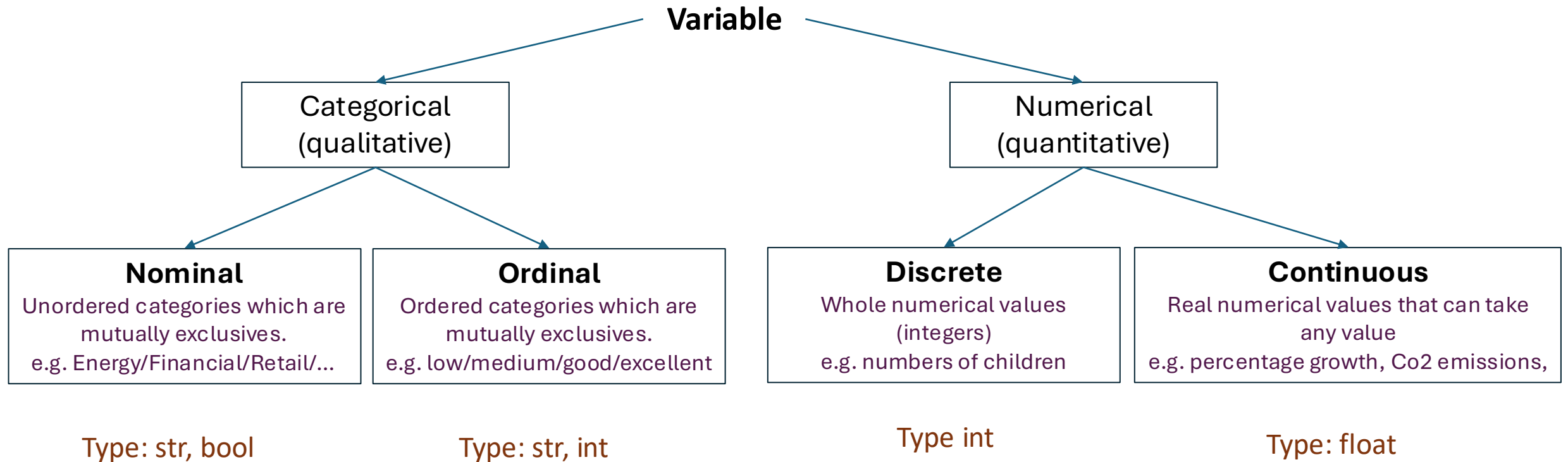
| Types of Data

In programming we spoke about data types. In statistics we speak about **types of data** (of type of variable). Although related, they are not the same !

The type of data determines **which summaries, visualizations, and statistical methods are appropriate**. The main types of data are:

- □ **Categorical (Nominal):** labels with no natural order (e.g., country, industry, color)
- □ **Categorical (Ordinal):** categories with a meaningful order (e.g., satisfaction rating, education level)
- □ **Binary:** two possible categories (e.g., Yes/No, Success/Failure)
- □ **Discrete Numerical:** countable numerical values (e.g., number of customers, number of defaults)
- □ **Continuous Numerical:** measurements that can take any value within an interval (e.g., return, height, weight)

| Types of Data and Their Correspondence to Python Types



These correspondences are approximate. Programming types describe how values are stored, whereas statistical types describe how variables should be analyzed.

What data looks like

Most datasets are organized as tables.

(also called **features**, **factors**, **attributes**, **predictors**, **covariates**, ...)

Variables

Records or
Data points

id_invoice	amount	age customer	items bought	credit card	Satisfaction_level
1	163.5	43	7	0	good
2	138.8	39	5	0	very good
3	175.9	48	7	1	low
4	157.5	45	11	1	low
5	600	43	2	0	very good
6	132.8	32	5	0	good
7	165.7	42	3	1	very good
8	134.1	39	2	1	good
9	174	39	1	1	very good
10	183.7	41	11	1	medium
11	157.8	39	17	1	medium
12	157.4	37	11	1	very good
13	154.3	43	16	1	low
14	169.2	37	3	1	very good
15	162.8	44	12	1	very good
16	108.4	38	6	1	good

Type of data
Possible data type

Categorical
Int, str

Continuous
float

Discrete/Continuous
Int, float

Discrete
int

Categorical
Bool, int

Ordinal
str, int

Visualizing Data

Different types of data require different visualizations.

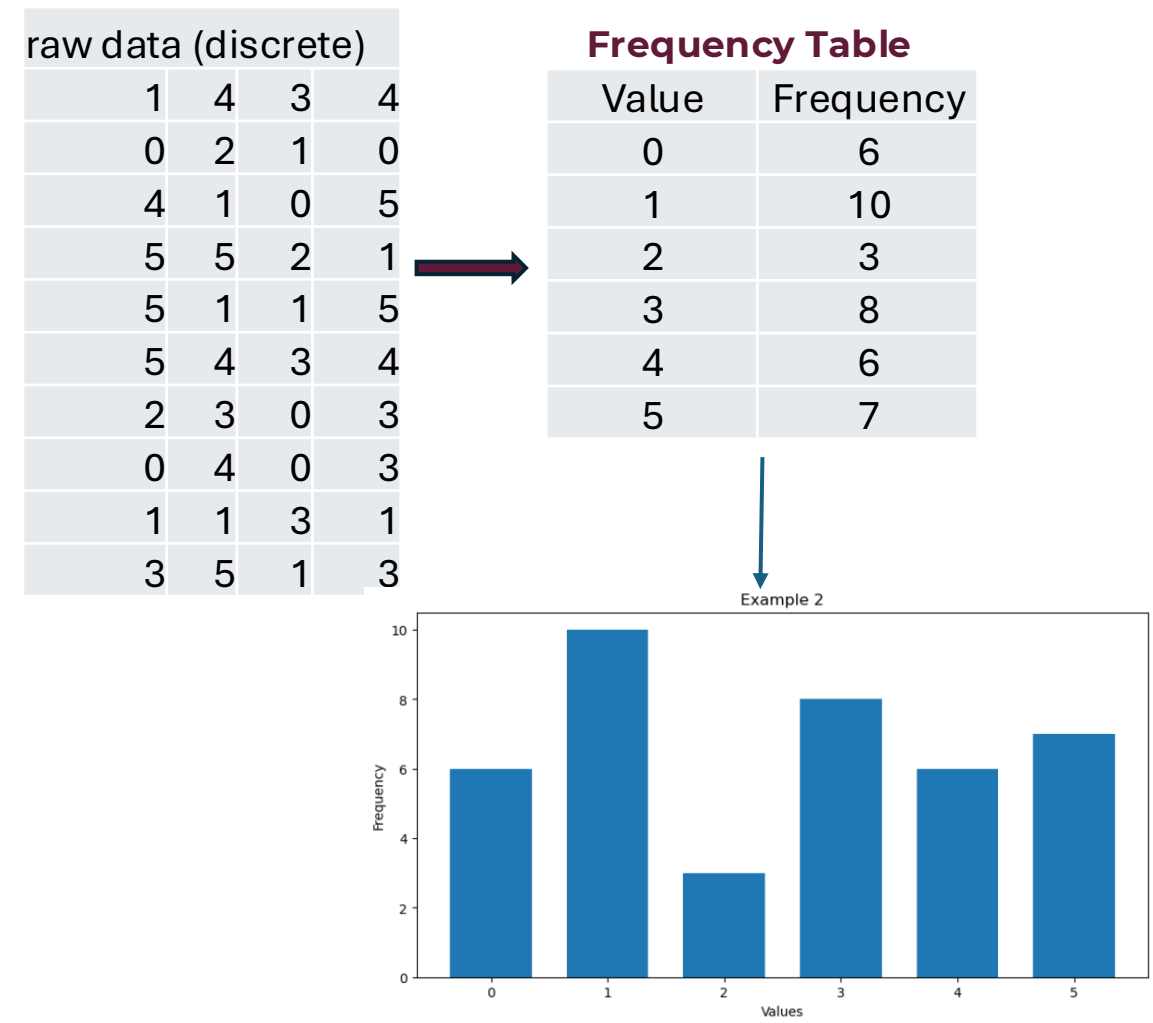
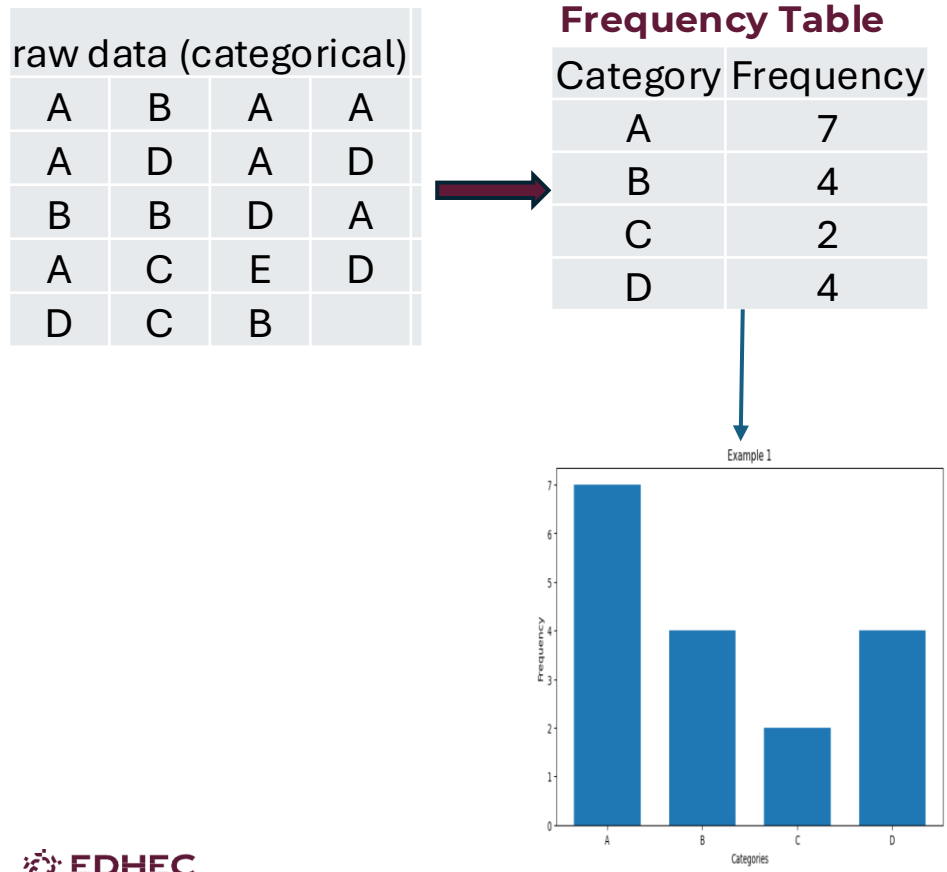
You need tabulate data

Type data / Type chart	Pie Chart	Line Chart	Bar Chart	Histogram	Scatter Plot (2 variables or more)
Categorical	X		X		
Ordinal	X		X		
Discrete		X	X	X	
Continuous (1 variable)		X		X	
Continuous (2 variables)				X	X

This table gives common choices, but the best graph depends on the question and context.

From Raw Data to Frequency Tables

Frequency tables transform raw observations into a concise summary by reporting how many times each category or value occurs.



Histograms for Continuous Data

For continuous variables, there are often too many distinct values to tabulate individually. Instead, we **group observations into intervals (called bins)** and count how many observations fall within each interval.

Raw Data (500 continuous numbers between 0 and 10)

0.3 9.26 6.18 1.83 4.09 1.64 3.72 7.05 0.81 7.24 7.8 2.07 3.8 8.71 7.95 0.98 0.66 5.83 3.04 3.29
6.58 6.08 0.3 8.81 4.86 4.59 3.22 0.97 0.49 6.71 6.36 7.82 9.89 1.83 4.91 6.68 1.03 8.09 3.4 8.98
6.49 4.23 6.43 3.86 1.88 7.06 0.11 2.7 2.93 4.75 8.3 1.55 6.31 1.59 5.39 4 6.63 2.77 1.17 5.23
7.38 2.52 5.26 3.32 3.82 9.71 3.75 8.9 8.97 0.6 6.14 1.29 8 9.28 0.29 4.91 5.35 4.84 3.2 0.9
4.12 3.3 0.13 4 5.37 9.4 7.71 8.8 8.51 6.41 2.1 4.96 6.43 4.2 4.45 1.78 8.62 1.85 9.47 8.52
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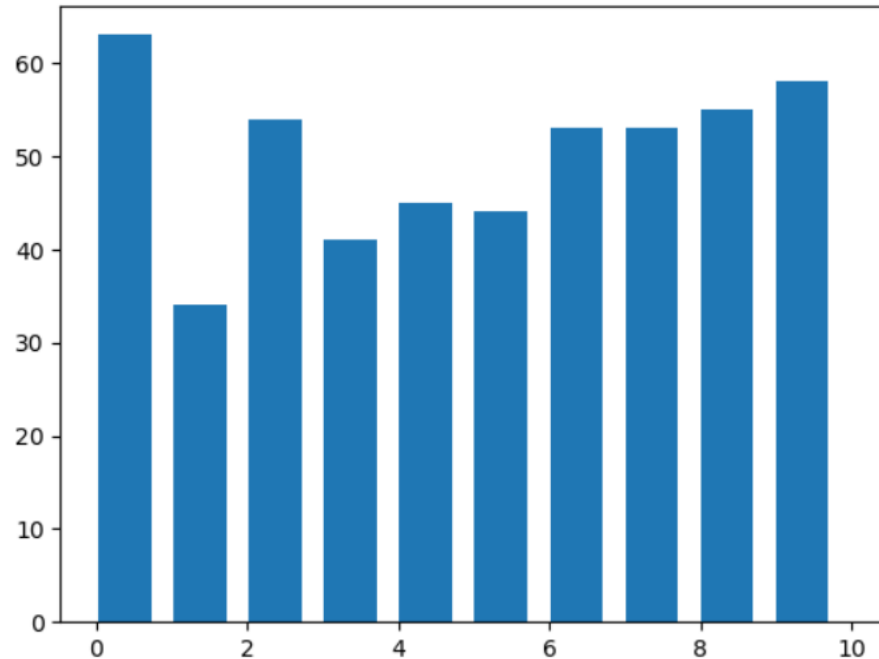
Frequency table with tabulated data

Range (exclude upper limit)	frequency
between 0 and 1	58
between 1 and 2	48
between 2 and 3	44
between 3 and 4	57
between 4 and 5	56
between 5 and 6	48
between 6 and 7	44
between 7 and 8	37
between 8 and 9	59
between 9 and 10	49

Total: 500 values

Histograms for Continuous Data

A **histogram** is the graphical representation of a grouped frequency table for a numerical variable.



Rule of thumb:

Categorical data → **Bar chart**

Numerical data → **Histogram**

Frequency table with tabulated data

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between 8 and 9	59
between 9 and 10	49

Total: 500 values